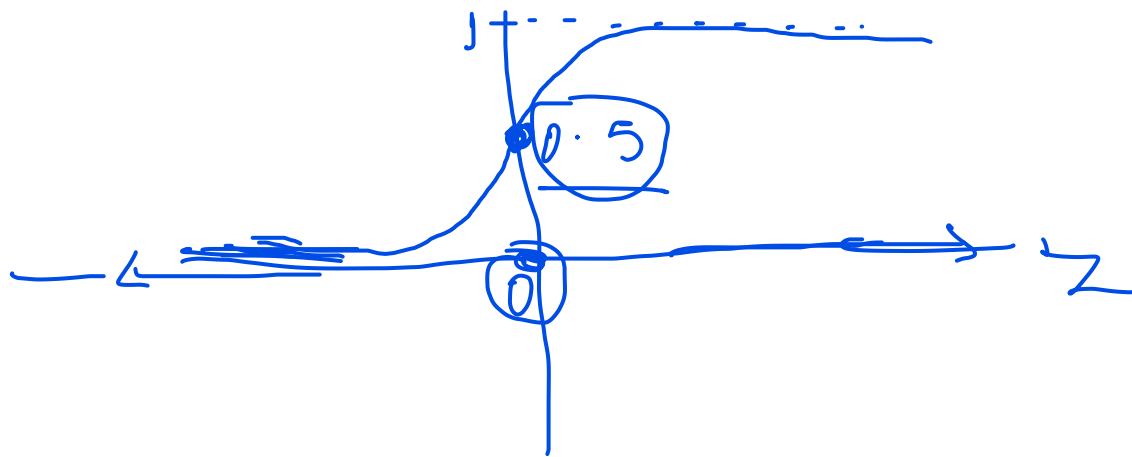


$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$0 \leq \sigma(Z) \leq 1$$



$$\frac{1}{1+e^0} = \frac{1}{1+e^{-100}}$$

$$\frac{1}{1+1} \rightarrow \boxed{0.5} \quad \frac{1}{1+e^{-100}} = \frac{1}{1+e^{-10}} =$$

$$\frac{1}{1+e^{100}} \approx \frac{1}{1+10^2} = \frac{1}{10^2}$$

$$\frac{1}{1 + \frac{1}{e^{10}}} = \frac{1}{1 + \frac{1}{22026}} \approx 1$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$z = \underline{\omega^T x + b} \propto \underline{\omega x + b}$$

$$\underline{\omega_0 + \omega_1 x}$$

$$f(x) = \frac{1}{1+e^{-x}}$$

$$\omega_0 + \omega_1 x_1 + \omega_2 x_2 + \dots$$

$$\underline{\sigma'(z)} = \underline{\sigma(z)(1-\sigma(z))} \quad \text{H.O} \quad 100\%$$

$$\text{Loss (Cross Entropy)} = - \left[\underbrace{y \log(\hat{y})}_{\substack{\text{Actual} \\ \text{Predicted}}} + \underbrace{(1-y) \log(1-\hat{y})}_{\substack{\text{Actual} \\ \text{Predicted}}} \right]$$

$$\hat{y} = w^T x + b$$

$$y = 0 \Rightarrow - [0 \log(\hat{y}) + (1-0) \log(1-\hat{y})] = \underline{\underline{-\log(1-\hat{y})}}$$

$$y = 1 \Rightarrow - [1 \log(\hat{y}) + \underline{(1-1)} \log(1-\hat{y})] = -\log(\hat{y})$$

$$Loss = - [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$Loss = - [y \log(\underline{w^T x + b}) + (1-y) \log(1 - \underline{w^T x + b})]$$

$$\frac{\partial L}{\partial w} = - \left[y \frac{1}{w^T x + b} [x] + (1-y) \frac{1}{1 - (w^T x + b)} [x] \right]$$

$$= - \left[\frac{y}{w^T x + b} + (1-y) \frac{1}{1 - (w^T x + b)} \right] x$$

$$= (\hat{y} - y) x$$

$$= (w^T x + b - y) x //$$

$$\text{softmax} =$$

$$\begin{bmatrix} 0.8 \\ 0.5 \\ 0.1 \end{bmatrix}$$

$$\frac{e^{\boxed{z_i}}}{\sum_{i=1}^n e^{z_i}}$$

=

$$\frac{e^{0.8}}{e^{0.8} + e^{0.5} + e^{0.1}} \rightarrow \boxed{}$$

$$+ \frac{e^{0.5}}{e^{0.8} + e^{0.5} + e^{0.1}} = \boxed{}$$

$$+ \frac{e^{0.1}}{e^{0.8} + e^{0.5} + e^{0.1}} = \boxed{}$$

$$\frac{}{1}$$

